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Modeling and Discrete Simulation

Subject 5

Queuing Models

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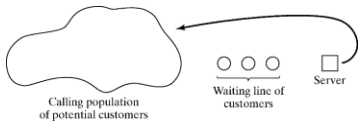


- Purpose
- General characteristics of queues
- Queuing systems
- Steady-State of Markovian Model
- Network of Queues

Slides are derived from the course textbook "Banks, Carson, Nelson, Nicol. Discrete-Event System Simulation, 5th Edition, Prentice Hall, 2010" and from textbook's course slides.

Purpose

- Simulation is often used in the analysis of queuing models.
- A simple but typical queuing model:



- Queuing models provide the analyst with a powerful tool for designing and evaluating the performance of queuing systems.
- Typical measures of system performance:
  - Server utilization, length of waiting lines, and delays of customers
  - For relatively simple systems, compute mathematically
  - For realistic models of complex systems, simulation is usually required.

Outline

- Discuss some well-known models (not development of queuing theories):
  - General characteristics of queues,
  - Meanings and relationships of important performance measures,
  - Estimation of mean measures of performance.
  - Effect of varying input parameters,
  - Mathematical solution of some basic queuing models.

CHARACTERISTICS OF QUEUING SYSTEMS

Characteristics of Queuing Systems

- Key elements of queuing systems:
  - **Customer**: refers to anything that arrives at a facility and requires service, e.g., people, machines, trucks, emails, cases, orders.
  - **Server**: refers to any resource that provides the requested service, e.g., repairpersons, retrieval machines, runways at airport, medical personnel, cranes, automatic packers, order pickers, CPUs, washing machines.

Calling Population

- **Calling population**: the population of potential customers, may be assumed to be finite or infinite.
  - Finite population model: if arrival rate depends on the number of customers being served and waiting, e.g., model of one corporate jet, if it is being repaired, the repair arrival rate becomes zero.
  - Infinite population model: if arrival rate is not affected by the number of customers being served and waiting, e.g., systems with large population of potential customers.
- In systems with large finite populations, it is generally safe to use infinite population models.
- The main difference between finite and infinite population models is how the arrival rate is defined (as given in above definitions).

System Capacity

- **System Capacity**: a limit on the number of customers that may be in the waiting line or system.
  - Limited capacity, e.g., an automatic car wash only has room for *10* cars to wait in line to enter the mechanism.
  - Unlimited capacity, e.g., concert ticket sales with no limit on the number of people allowed to wait to purchase tickets.
- When a system has limited capacity, a distinction is made between the *arrival rate* (the number of arrivals per unit time) and the *effective arrival rate* (the number who arrive and enter the system per unit time).

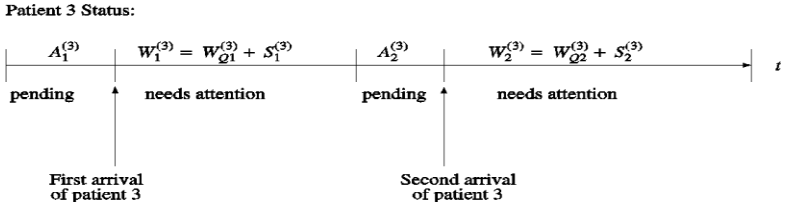
Arrival Process

- For infinite-population models:
  - It is characterized in terms of interarrival times of successive customers.
  - **Random arrivals:** interarrival times usually characterized by a probability distribution.
    - Most important model: Poisson arrival process (with rate λ), where A_n represents the interarrival time between customer *n-1* and customer *n*, and is exponentially distributed (with mean $1/\lambda$).
  - **Scheduled arrivals:** interarrival times can be constant or constant plus or minus a small random amount to represent early or late arrivals.
    - e.g., patients to a physician or scheduled airline flight arrivals to an airport.
  - **At least one customer is assumed to always be present**, so the server is never idle, e.g., sufficient raw material for a machine.

Arrival Process

- For finite-population models:
  - Customer is pending when the customer is outside the queuing system, e.g., machine-repair problem: a machine is "pending" when it is operating, it becomes "not pending" the instant it demands service from the repairman.
  - Runtime of a customer is the length of time from departure from the queuing system until that customer's next arrival to the queue, e.g., machine-repair problem, machines are customers and a runtime is time to failure.
  - Let $A_1^{(i)}, A_2^{(i)}, \ldots$ be the successive runtimes of customer *i*; and $S_1^{(i)}, S_2^{(i)}$ be the corresponding successive system times, and $W_{Q1}^{(i)}, W_{Q2}^{(i)}, \ldots$ be the successive waiting times for service to begin, then:

$$W_n^{(i)} = W_{Qn}^{(i)} + S_n^{(i)}$$



Queue Behavior and Queue Discipline

- **Queue behavior:** the actions of customers while in a queue waiting for service to begin, for example:
  - Balk: leave when they see that the line is too long,
  - Renege: leave after being in the line when its moving too slowly,
  - Jockey: move from one line to a shorter line.

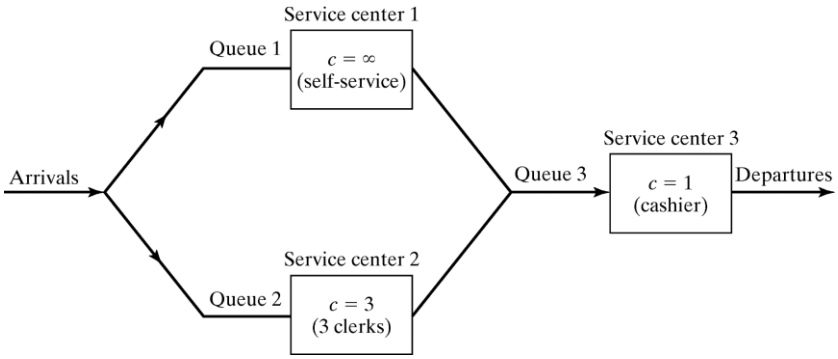
- **Queue discipline:** the logical ordering of customers in a queue that determines which customer is chosen for service when a server becomes free, for example:
  - First-in-first-out (FIFO)
  - Last-in-first-out (LIFO)
  - Service in random order (SIRO)
  - Shortest processing time first (SPT)
  - Service according to priority (PR).

Service Times and Service Mechanism

- **Service times** of successive arrivals are denoted by S_1, S_2, S_3.
  - May be constant or random.
  - $\{S_1, S_2, S_3, \ldots\}$ is usually characterized as a sequence of independent and identically distributed random variables, e.g., exponential, Weibull, gamma, lognormal, and truncated normal distribution.
- A queueing system consists of a number of service centers and interconnected queues.
  - Each service center consists of some number of servers, c, working in parallel, upon getting to the head of the line, a customer takes the 1st available server.

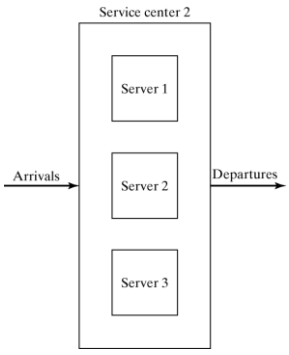
Service Times and Service Mechanism

- Example: consider a discount warehouse where customers may:
  - Serve themselves before paying at the cashier:



Service Times and Service Mechanism

  - Wait for one of the three clerks:



  - Other possible variations:
    - Batch service (a server serving several customers simultaneously), or
    - customer requires several servers simultaneously.

Queuing Notation

- A notation system (*Kendall's Notation*) for parallel server queues: *A/B/c/N/K*
  - *A* represents the interarrival-time distribution,
  - *B* represents the service-time distribution,
  - *c* represents the number of parallel servers,
  - *N* represents the system capacity,
  - *K* represents the size of the calling population.

e.g. **M/M/1/∞/ ∞, M/M/2, M/M/1/10, M/M/2/50/50, M/D/2, G/G/1…**

Queuing Notation

- Primary performance measures of queuing systems:
  - P_n: steady-state probability of having n customers in system,
  - $P_n(t)$: probability of n customers in system at time t,
  - λ: arrival rate,
  - λ_e: effective arrival rate,
  - μ: service rate of one server,
  - ρ : server utilization,
  - A_n: interarrival time between customers n-1 and n,
  - S_n: service time of the nth arriving customer,
  - W_n: total time spent in system by the nth arriving customer,
  - W_n^Q: total time spent in the waiting line by customer n,
  - $L(t)$: the number of customers in system at time t,
  - $L_Q(t)$: the number of customers in queue at time t,
  - L: long-run time-average number of customers in system,
  - L_Q: long-run time-average number of customers in queue,
  - w: long-run average time spent in system per customer,
  - w_Q: long-run average time spent in queue per customer.

Time-Average Number in System, *L*

- Consider a queuing system over a period of time *T*,
  - Let T_i denote the total time during [*0,T*] in which the system contained exactly *i* customers, the time-weighted-average number in a system is defined by:

$$\hat{L} = \frac{1}{T}\sum_{i=0}^{\infty} iT_i = \sum_{i=0}^{\infty} i\left(\frac{T_i}{T}\right)$$

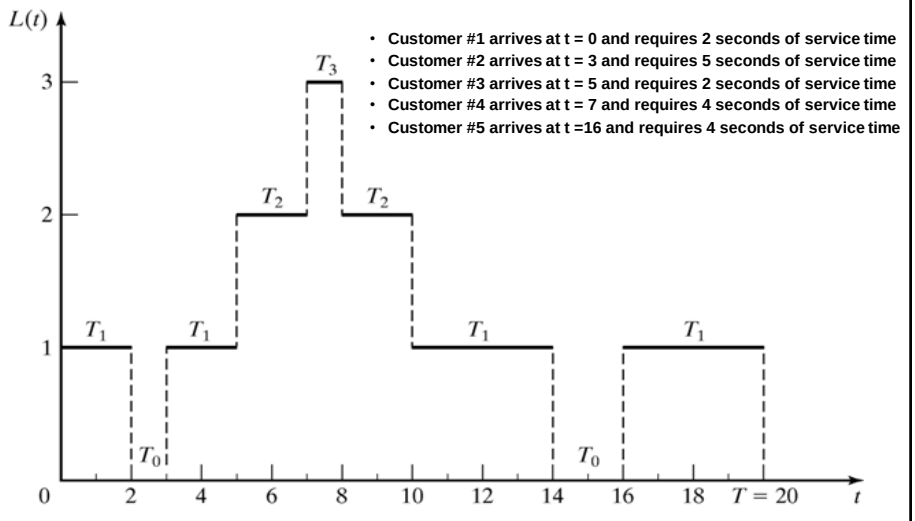
  - Consider the total area under the function *L(t)*, then,

$$\hat{L} = \frac{1}{T}\sum_{i=0}^{\infty} iT_i = \frac{1}{T}\int_0^T L(t)dt$$

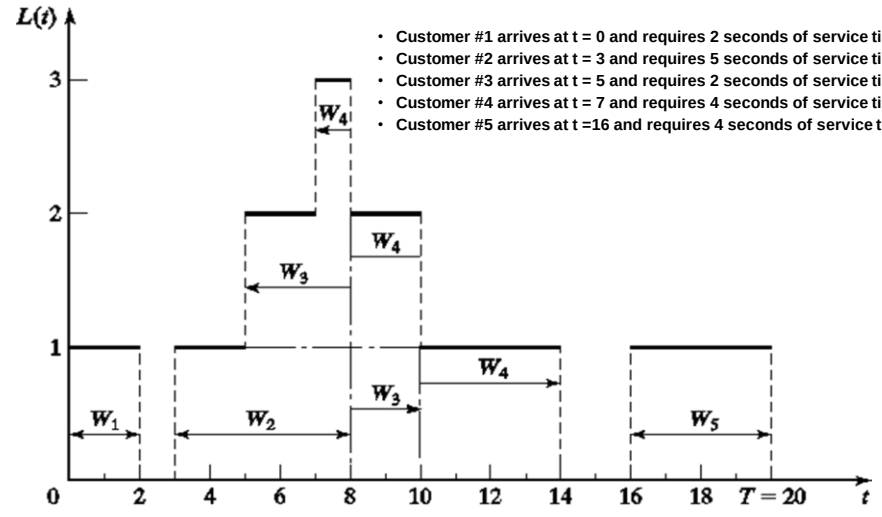
  - The long-run time-average # in system, with probability *1*:

$$\hat{L} = \frac{1}{T}\int_0^T L(t)dt \to L \quad \text{as} \quad T \to \infty$$

Time-Average Number in System, *L*

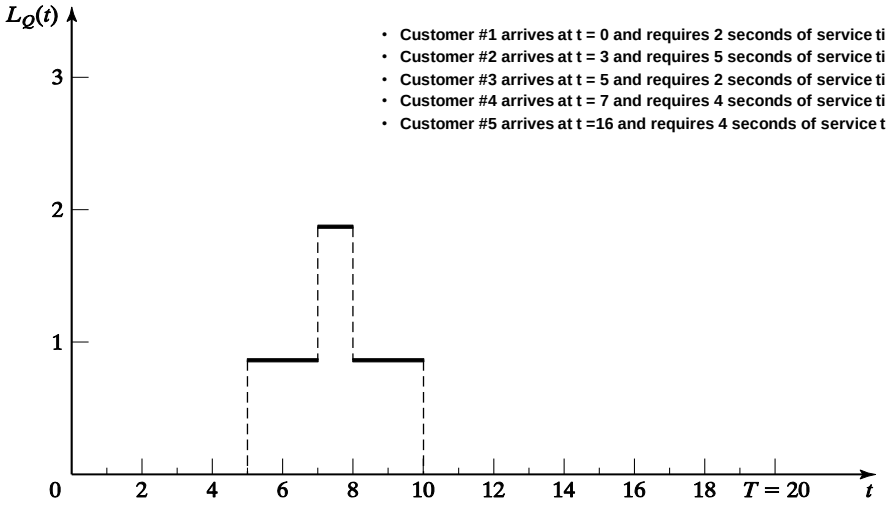


Average Time Spent in System Per Customer *w*



- Customer #1 arrives at t = 0 and requires 2 seconds of service time
- Customer #2 arrives at t = 3 and requires 5 seconds of service time
- Customer #3 arrives at t = 5 and requires 2 seconds of service time
- Customer #4 arrives at t = 7 and requires 4 seconds of service time
- Customer #5 arrives at t =16 and requires 4 seconds of service time

Average Time Spent in System Per Customer *w*



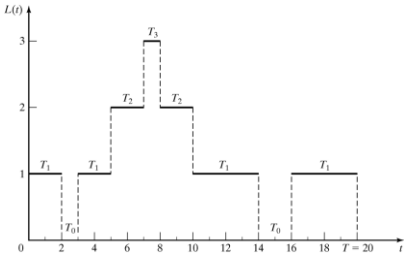
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- Customer #5 arrives at t =16 and requires 4 seconds of service time

Time-Average Number in System, *L*

– The time-weighted-average number in queue is:

$$\hat{L}_Q = \frac{1}{T}\sum_{i=0}^{\infty} iT_i^Q = \frac{1}{T}\int_0^T L_Q(t)dt \to L_Q \quad \text{as} \quad T \to \infty$$

– *G/G/1/N/K* example: consider the results from the queueing system ($N > 4, K > 3$).



$$\hat{L} = [0(3) + 1(12) + 2(4) + 3(1)]/20$$
$$= 23/20 = 1.15 \text{ customers}$$

$$L_Q(t) = \begin{cases} 0, & \text{if } L(t) = 0 \\ L(t) - 1, & \text{if } L(t) \geq 1 \end{cases}$$

$$\hat{L}_Q = \frac{0(15) + 1(4) + 2(1)}{20} = 0.3 \text{ customers}$$

Average Time Spent in System Per Customer, *w*

- The average time spent in system per customer, called the average system time, is:

$$\hat{w} = \frac{1}{N}\sum_{i=1}^{N} W_i$$

where $W_1, W_2, \ldots, W_N$ are the individual times that each of the N customers spend in the system during $[0,T]$.

– For stable systems: $\hat{w} \to w$ as $N \to \infty$

– If the system under consideration is the queue alone:

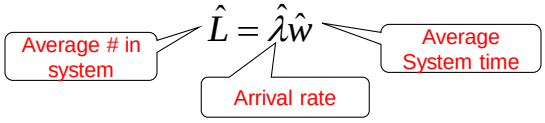
$$\hat{w}_Q = \frac{1}{N}\sum_{i=1}^{N} W_i^Q \to w_Q \quad \text{as} \quad N \to \infty$$

– *G/G/1/N/K* example (cont.): the average system time is

$$\hat{w} = \frac{W_1 + W_2 + \ldots + W_5}{5} = \frac{2 + (8-3) + \ldots + (20-16)}{5} = 4.6 \text{ time units}$$

The Conservation Equation

- Conservation equation (a.k.a. Little's law)



$$L = \lambda w \quad \text{as} \quad T \to \infty \text{ and } N \to \infty$$

  - Holds for almost all queuing systems or subsystems (regardless of the number of servers, the queue discipline, or other special circumstances).
  - *G/G/1/N/K* example (cont.): On average, one arrival every *4* time units and each arrival spends *4.6* time units in the system. Hence, at an arbitrary point in time, there is *(1/4)(4.6) = 1.15* customers present on average.

Server Utilization

- Definition: the proportion of time that a server is busy.
  - Observed server utilization, $\hat{\rho}$, is defined over a specified time interval [0,T].
  - Long-run server utilization is ρ.
  - For systems with long-run stability: $\hat{\rho} \to \rho$ as $T \to \infty$

Server Utilization

- For *G/G/1/∞/∞* queues:
  - Any single-server queuing system with average arrival rate λ customers per time unit, where average service time $E(S) = 1/\mu$ time units, infinite queue capacity and calling population.
  - Conservation equation, $L = \lambda w$, can be applied.
  - For a stable system, the average arrival rate to the server, λ_s, must be identical to λ.
  - The average number of customers in the server is:

$$\hat{L}_s = \frac{1}{T}\int_0^T \left(L(t) - L_Q(t)\right)dt = \frac{T - T_0}{T}$$

Server Utilization

  - In general, for a single-server queue:

$$\hat{L}_s = \hat{\rho} \rightarrow L_s = \rho \quad \text{as } T \rightarrow \infty$$

$$\text{and} \quad \rho = \lambda E(s) = \frac{\lambda}{\mu}$$

- For a single-server stable queue: $\rho = \frac{\lambda}{\mu} < 1$
- For an unstable queue ($\lambda > \mu$), long-run server utilization is *1*.

Server Utilization

- For $G/G/c/\infty/\infty$ queues:
  - A system with c identical servers in parallel.
  - If an arriving customer finds more than one server idle, the customer chooses a server without favoring any particular server.
  - For systems in statistical equilibrium, the average number of busy servers, L_s, is: $L_s = \lambda E(s) = \lambda/\mu$.
  - The long-run average server utilization is:

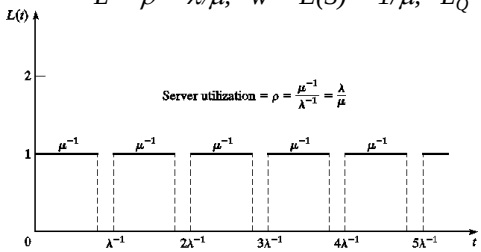
$$\rho = \frac{L_s}{c} = \frac{\lambda}{c\mu}, \quad \text{where } \lambda < c\mu \text{ for stable systems}$$

  - The utilization, ρ, can be interpreted as the proportion of time an arbitrary server is busy in the long run.

Server Utilization and System Performance

- System performance varies widely for a given utilization ρ.
  - For example, a *D/D/1* queue where $E(A) = 1/\lambda$ and $E(S) = 1/\mu$, where:

$$L = \rho = \lambda/\mu, \quad w = E(S) = 1/\mu, \quad L_Q = W_Q = 0.$$



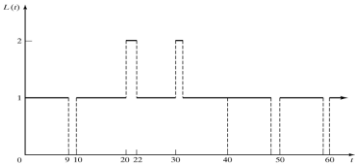
  - By varying λ and μ, server utilization can assume any value between *0* and *1*.
  - Yet there is never any waiting line.
  - In general, variability of interarrival and service times causes waiting lines to fluctuate in length.

Server Utilization and System Performance

- Example: A physician who schedules patients every *10* minutes and spends S_i minutes with the ith patient:

$$S_i = \begin{cases} 9 \text{ minutes with probability } 0.9 \\ 12 \text{ minutes with probability } 0.1 \end{cases}$$

  - Arrivals are deterministic, $A_1 = A_2 = \ldots = \lambda^{-1} = 10$.
  - Services are stochastic, $E(S_i) = 9.3$ min and $V(S_0) = 0.81$ min^2.
  - On average, the physician's utilization $= \rho = \lambda/\mu = 0.93 < 1$.
  - Consider the system is simulated with service times: $S_1 = 9$, $S_2 = 12$, $S_3 = 9$, $S_4 = 9$, $S_5 = 9$, ... The system becomes:



  - The occurrence of a relatively long service time ($S_2 = 12$) causes a waiting line to form temporarily.

Costs in Queuing Problems

- Costs can be associated with various aspects of the waiting line or servers:
  - System incurs a cost for each customer in the queue, say at a rate of *$10* per hour per customer.
    - The average cost per customer is:

$$\sum_{j=1}^{N} \frac{\$10 * W_j^Q}{N} = \$10 * \hat{w}_Q$$

(W_j^Q is the time customer j spends in queue)

    - If $\hat{\lambda}$ customers per hour arrive (on average), the average cost per hour is:

$$\left(\hat{\lambda}\frac{\text{customer}}{\text{hour}}\right)\left(\frac{\$10 * \hat{w}_Q}{\text{customer}}\right) = \$10 * \hat{\lambda}\hat{w}_Q = \$10 * \hat{L}_Q \text{ / hour}$$

  - Server may also impose costs on the system, if a group of c parallel servers ($1 \le c \le \infty$) have utilization ρ, each server imposes a cost of *$5* per hour while busy.
    - The total server cost is: $\$5 * c\rho$.

STEADY-STATE OF MARKOVIAN MODEL

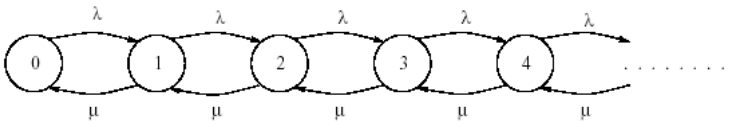
Steady-State Behavior of Infinite-Population Markovian Models

- **Markovian models:** exponential-distribution arrival process (mean arrival rate = λ).
- Service times may be exponentially distributed as well (*M*) or arbitrary (*G*).
- A queuing system is in *statistical equilibrium* or *steady state* if the probability that the system is in a given state is not time dependent:

$$P(L(t) = n) = P_n(t) = P_n.$$

- Mathematical models in this lecture can be used to obtain approximate results even when the model assumptions do not strictly hold (as a rough guide).
- Simulation can be used for more refined analysis (more faithful representation for complex systems).

Steady-State Behavior of Infinite-Population Markovian Models



CTMC for the M/M/1 queue

Steady-State Behavior of Infinite-Population Markovian Models

- For the simple model studied in this chapter, the steady-state parameter, L, the time-average number of customers in the system is:

$$L = \sum_{n=0}^{\infty} nP_n$$

  - Apply Little's equation to the whole system and to the queue alone:

$$w = \frac{L}{\lambda}, \quad w_Q = w - \frac{1}{\mu}$$

$$L_Q = \lambda w_Q$$

- *M/G/c/∞/∞* example: to have a statistical equilibrium, a necessary and sufficient condition is $\lambda/(c\mu) < 1$.

M/G/1 Queues

- Single-server queues with Poisson arrivals & unlimited capacity.
- Suppose service times have mean $1/\mu$ and variance σ^2 and $\rho = \lambda/\mu < 1$, the steady-state parameters of *M/G/1* queue:

$$\rho = \lambda/\mu, \quad P_0 = 1-\rho$$

$$L = \rho + \frac{\rho^2(1+\sigma^2\mu^2)}{2(1-\rho)}, \quad L_Q = \frac{\rho^2(1+\sigma^2\mu^2)}{2(1-\rho)}$$

$$w = \frac{1}{\mu} + \frac{\lambda(1/\mu^2+\sigma^2)}{2(1-\rho)}, \quad w_Q = \frac{\lambda(1/\mu^2+\sigma^2)}{2(1-\rho)}$$

M/G/1 Queues

- No simple expression for the steady-state probabilities P_0, P_1, …
- $L - L_Q = \rho$ is the time-average number of customers being served.
- Average length of queue, L_Q, can be rewritten as:

$$L_Q = \frac{\rho^2}{2(1-\rho)} + \frac{\lambda^2\sigma^2}{2(1-\rho)}$$

  - If λ and μ are held constant, L_Q depends on the variability, σ^2, of the service times.

M/G/1 Queues

- Example: Two workers competing for a job, Able claims to be faster than Baker on average, but Baker claims to be more consistent,
  - Poisson arrivals at rate $\lambda = 2$ per hour ($1/30$ per minute).
  - Able: $1/\mu = 24$ minutes and $\sigma^2 = 20^2 = 400$ minutes^2:

$$L_Q = \frac{(1/30)^2[24^2 + 400]}{2(1-4/5)} = 2.711 \text{ customers}$$

    - The proportion of arrivals who find Able idle and thus experience no delay is $P_0 = 1-\rho = 1/5 = 20\%$.
  - Baker: $1/\mu = 25$ *minutes* and $\sigma^2 = 2^2 = 4$ minutes^2:

$$L_Q = \frac{(1/30)^2[25^2 + 4]}{2(1-5/6)} = 2.097 \text{ customers}$$

    - The proportion of arrivals who find Baker idle and thus experience no delay is $P_0 = 1-\rho = 1/6 = 16.7\%$.
  - Although working faster on average, Able's greater service variability results in an average queue length about *30%* greater than Baker's.

M/M/1 Queues

- Suppose the service times in an *M/G/1* queue are exponentially distributed with mean $1/\mu$, then the variance is $\sigma^2 = 1/\mu^2$.
  - *M/M/1* queue is a useful approximate model when service times have standard deviation approximately equal to their means.
  - The steady-state parameters:

$$\rho = \lambda/\mu, \quad P_n = (1-\rho)\rho^n$$

$$L = \frac{\lambda}{\mu-\lambda} = \frac{\rho}{1-\rho}, \quad L_Q = \frac{\lambda^2}{\mu(\mu-\lambda)} = \frac{\rho^2}{1-\rho}$$

$$w = \frac{1}{\mu-\lambda} = \frac{1}{\mu(1-\rho)}, \quad w_Q = \frac{\lambda}{\mu(\mu-\lambda)} = \frac{\rho}{\mu(1-\rho)}$$

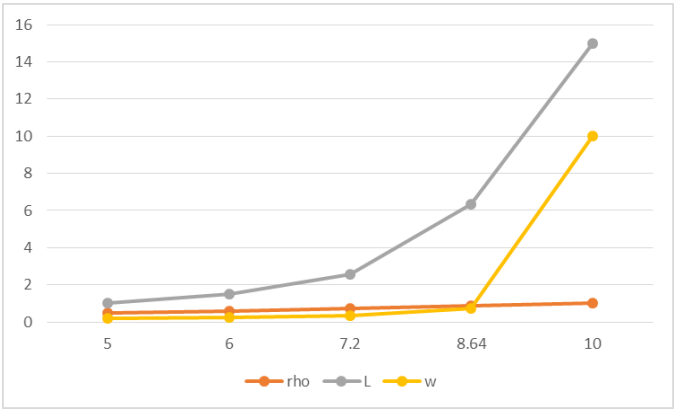
M/M/1 Queues

- Example: *M/M/1* queue with service rate $\mu=10$ customers per hour.
  - Consider how L and w increase as arrival rate, λ, increases from 5 to 8.64 by increments of 20%:

	λ	5.0	6.0	7.2	8.64	10.0
$\lambda/\mu \rightarrow$	ρ	0.500	0.600	0.720	0.864	1.000
$\rho/(1-\rho) \rightarrow$	L	1.00	1.50	2.57	6.35	∞
$L/\lambda \rightarrow$	w	0.20	0.25	0.36	0.73	∞

  - If $\lambda/\mu \geq 1$, waiting lines tend to continually grow in length.
  - Increase in average system time (w) and average number in system (L) is highly nonlinear as a function of ρ.

M/M/1 Queues



  - If $\lambda/\mu \geq 1$, waiting lines tend to continually grow in length.
  - For $\lambda/\mu < 1$, increase in average system time (w) and average number in system (L) is highly nonlinear as a function of ρ.

Effect of Utilization and Service Variability

- For almost all queues, if lines are too long, they can be reduced by decreasing server utilization (ρ) or by decreasing the service time variability (σ^2).
- A measure of the variability of a distribution, coefficient of variation (cv):

$$(cv)^2 = \frac{V(X)}{[E(X)]^2}$$

  - The larger cv is, the more variable is the distribution relative to its expected value

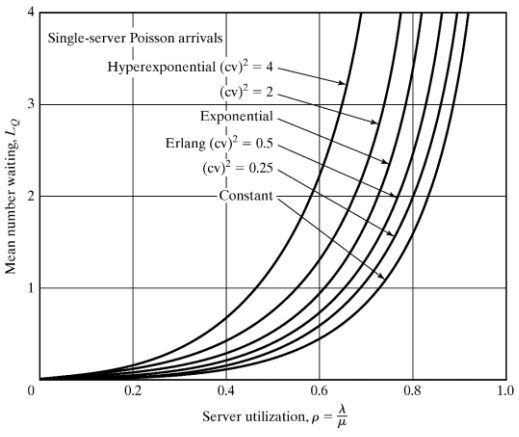
Effect of Utilization and Service Variability

- Consider L_Q for any *M/G/1* queue:

$$L_Q = \frac{\rho^2(1+\sigma^2\mu^2)}{2(1-\rho)} = \left(\frac{\rho^2}{1-\rho}\right)\left(\frac{1+(cv)^2}{2}\right)$$

L_Q for *M/M/1* queue

Corrects the *M/M/1* formula to account for a non-exponential service time dist'n



Multiserver Queue

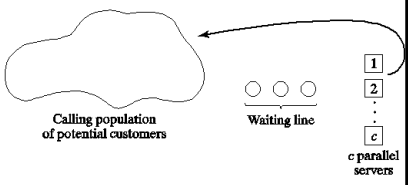
- *M/M/c/∞/∞* queue: *c* channels operating in parallel.
  - Each channel has an independent and identical exponential service-time distribution, with mean $1/\mu$.
  - To achieve statistical equilibrium, the offered load (λ/μ) must satisfy $\lambda/\mu < c$, where $\lambda/(c\mu) = \rho$ is the server utilization.
  - Some of the steady-state probabilities:

$$\rho = \lambda / c\mu$$

$$P_0 = \left\{ \left[\sum_{n=0}^{c-1} \frac{(\lambda/\mu)^n}{n!} \right] + \left[\left(\frac{\lambda}{\mu} \right)^c \left(\frac{1}{c!} \right) \left(\frac{c\mu}{c\mu - \lambda} \right) \right] \right\}^{-1}$$

$$L = c\rho + \frac{(c\rho)^{c+1} P_0}{c(c!)(1-\rho)^2} = c\rho + \frac{\rho\, P(L(\infty) \geq c)}{1-\rho}$$

$$w = \frac{L}{\lambda}$$



Multiserver Queue

- Other common multiserver queueing models:
  - *M/G/c/∞*: general service times and c parallel server. The parameters can be approximated from those of the *M/M/c/∞/∞* model.
  - *M/G/∞:* general service times and infinite number of servers, e.g., customer is its own system, service capacity far exceeds service demand.
  - *M/M/C/N/∞*: service times are exponentially distributed at rate 1/μ and c servers where the total system capacity is $N \geq c$ customer (when an arrival occurs and the system is full, that arrival is turned away).

Steady-State Behavior of Finite-Population Models

- When the calling population is small, the presence of one or more customers in the system has a strong effect on the distribution of future arrivals.
- Consider a finite-calling population model with K customers ($M/M/c/K/K$):
  - The time between the end of one service visit and the next call for service is exponentially distributed, (mean = $1/\lambda$).
  - Service times are also exponentially distributed.
  - c parallel servers and system capacity is K.

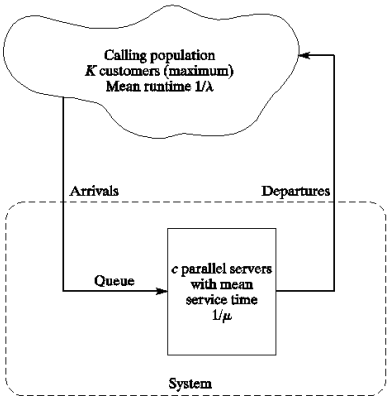
Steady-State Behavior of Finite-Population Models

- Some of the steady-state probabilities:

$$P_0 = \left\{ \sum_{n=0}^{c-1} \binom{K}{n} \left(\frac{\lambda}{\mu}\right)^n + \sum_{n=c}^{K} \frac{K!}{(K-n)!c!c^{n-c}} \left(\frac{\lambda}{\mu}\right)^n \right\}^{-1}$$

$$P_n = \begin{cases} \binom{K}{n} \left(\frac{\lambda}{\mu}\right)^n P_0, & n = 0,1,\ldots,c-1 \\ \frac{K!}{(K-n)!c!c^{n-c}} \left(\frac{\lambda}{\mu}\right)^n, & n = c, c+1,\ldots K \end{cases}$$

$$L = \sum_{n=0}^{K} nP_n, \quad w = L/\lambda_e, \quad \rho = \lambda_e / c\mu$$



where λ_e is the long run effective arrival rate of customers to queue (or entering / exiting service)

$$\lambda_e = \sum_{n=0}^{K} (K-n)\lambda P_n$$

Steady-State Behavior of Finite-Population Models

- Example: two workers who are responsible for *10* milling machines.
  - Machines run on the average for *20* minutes, then require an average 5-minute service period, both times exponentially distributed: $\lambda = 1/20$ *and* $\mu = 1/5$.
  - All of the performance measures depend on P_0:

$$P_0 = \left\{ \sum_{n=0}^{2-1} \binom{10}{n} \left(\frac{5}{20}\right)^n + \sum_{n=2}^{10} \frac{10!}{(10-n)!2!2^{n-2}} \left(\frac{5}{20}\right)^n \right\}^{-1} = 0.065$$

    - Then, we can obtain the other P_n.
    - Expected number of machines in system:

$$L = \sum_{n=0}^{10} nP_n = 3.17 \text{ machines}$$

    - The average number of running machines:

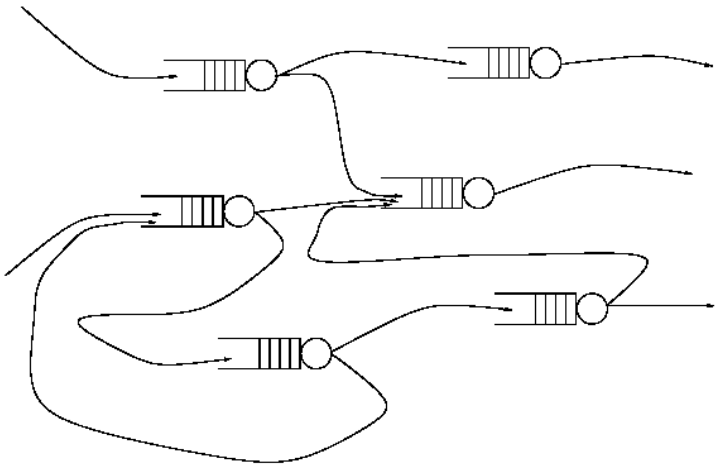
$$K - L = 10 - 3.17 = 6.83 \text{ machines}$$

NETWORKS OF QUEUES

Networks of Queues

- Many systems are naturally modeled as **networks of single queues**: customers departing from one queue may be routed to another.
- The following results assume a stable system with infinite calling population and no limit on system capacity:
  - Provided that no customers are created or destroyed in the queue, then the departure rate out of a queue is the same as the arrival rate into the queue (over the long run).
  - If customers arrive to queue i at rate λ_i, and a fraction $0 \le p_{ij} \le 1$ of them are routed to queue j upon departure, then the arrival rate from queue i to queue j is $\lambda_i p_{ij}$ (over the long run).

Networks of Queues



An Open Queueing Network

Networks of Queues

- The overall arrival rate into queue j:

$$\lambda_j = a_j + \sum_{\text{all } i} \lambda_i p_{ij}$$

Arrival rate from outside the network

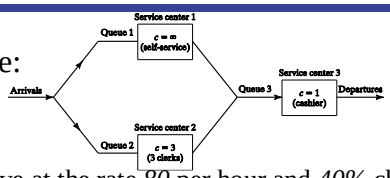
Sum of arrival rates from other queues in network

- If queue j has $c_j < \infty$ parallel servers, each working at rate μ_j, then the long-run utilization of each server is $\rho_j = \lambda_j/(c\mu_j)$ (where $\rho_j < 1$ for stable queue).
- If arrivals from outside the network form a Poisson process with rate a_j for each queue j, and if there are c_j identical servers delivering exponentially distributed service times with mean $1/\mu_j$, then, in steady state, queue j behaves likes an $M/M/c_j$ queue with arrival rate

$$\lambda_j = a_j + \sum_{\text{all } i} \lambda_i p_{ij}$$

Networks of Queues

- Discount store example:



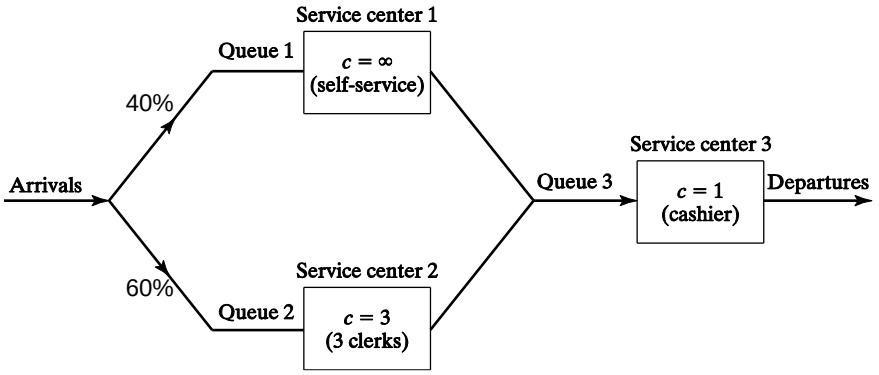
- Suppose customers arrive at the rate *80* per hour and *40%* choose self-service. Hence:
  - Arrival rate to service center 1 is $\lambda_1 = 80(0.4) = 32$ per hour
  - Arrival rate to service center 2 is $\lambda_2 = 80(0.6) = 48$ per hour.
- $c_2 = 3$ clerks and $\mu_2 = 20$ customers per hour.
- The long-run utilization of the clerks is:

$$\rho_2 = 48/(3*20) = 0.8$$

- All customers must see the cashier at service center 3, the overall rate to service center 3 is $\lambda_3 = \lambda_1 + \lambda_2 = 80$ per hour.
  - If $\mu_3 = 90$ per hour, then the utilization of the cashier is:

$$\rho_3 = 80/90 = 0.89$$

Networks of Queues



Summary

- Introduced basic concepts of queueing models.
- Show how simulation, and some times mathematical analysis, can be used to estimate the performance measures of a system.
- Commonly used performance measures: L, L_Q, w, w_Q, ρ, and λ_e.
- When simulating any system that evolves over time, analyst must decide whether to study transient behavior or steady-state behavior.
  - Simple formulas exist for the steady-state behavior of some queues.
- Simple models can be solved mathematically, and can be useful in providing a rough estimate of a performance measure.